

YULIN LUO

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👤 RESEARCH PROFILE

I am a Ph.D. student in Computer Science at Peking University, advised by Prof. Shanghang Zhang. My research studies how embodied foundation models can generalize in open-world environments, connecting model generalization, embodied data and benchmark construction, and model-data co-evolution.

🎓 EDUCATION

Peking University, Ph.D. in Computer Science Sep. 2023 – Present

- Advisor: Prof. Shanghang Zhang.

Shanghai Jiao Tong University, B.Eng. in Automation Sep. 2019 – Jun. 2023

- GPA: 3.87/4.30; Major rank: 2/89.

🔬 RESEARCH INTERESTS

- **Model:** building generalizable embodied foundation models for open-world robot intelligence.
- **Data / Benchmark:** constructing embodied-agent benchmarks, multimodal evaluations, and robot-scenario data.
- **Model-Data Co-evolution:** scaling embodied intelligence through data-driven model improvement and model-guided data construction.

📄 SELECTED PUBLICATIONS

- **RoboBench: A Comprehensive Evaluation Benchmark for MLLMs as Embodied Brain.**
Yulin Luo*, Chun-Kai Fan*, Menghang Dong*, Jiayi Shi*, Mengdi Zhao*, Bo-Wen Zhang*, et al., Shanghang Zhang.
European Conference on Computer Vision (ECCV), 2026. CCF-B. First author.
- **LLM as Dataset Analyst: Subpopulation Structure Discovery with Large Language Model.**
Yulin Luo*, Ruichuan An*, Bocheng Zou, Yiming Tang, Jiaming Liu, Shanghang Zhang.
European Conference on Computer Vision (ECCV), 2024. CCF-B. First author.
- **Look Before Acting: Enhancing Vision Foundation Representations for Vision-Language-Action Models.**
Yulin Luo*, Hao Chen*, Zhixuan Wu*, Bo Sui*, Jiaming Liu*, et al., Shanghang Zhang.
arXiv preprint, 2026. First author.
- **Decomposing the Neurons: Activation Sparsity via Mixture of Experts for Continual Test-Time Adaptation.**
Rongyu Zhang*, Aosong Cheng*, Yulin Luo*, Gaole Dai, Huanrui Yang, et al., Shanghang Zhang.
AAAI Conference on Artificial Intelligence (AAAI), 2026. CCF-A. Oral. Co-first author.
- **RoboMIND: Benchmark on Multi-Embodiment Intelligence Normative Data for Robot Manipulation.**
Kun Wu, Chengkai Hou, Jiaming Liu, et al., Yulin Luo, et al.
Robotics: Science and Systems (RSS), 2025. CCF-B.
- **RandStainNA: Learning Stain-Agnostic Features from Histology Slides by Bridging Stain Augmentation and Normalization.**
Yiqing Shen*, Yulin Luo*, Dinggang Shen, Jing Ke.
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2022. CCF-B. Co-first author.
- **Efficient Deweather Mixture-of-Experts with Uncertainty-Aware Feature-Wise Linear Modulation.**
Rongyu Zhang, Yulin Luo, Jiaming Liu, Huanrui Yang, et al., Shanghang Zhang.
AAAI Conference on Artificial Intelligence (AAAI), 2024. CCF-A.

RESEARCH INTERNSHIPS

Simplexity Robotics, Research Intern

Dec. 2025 – Present

- Research on embodied intelligence and generalizable robot foundation models.

Beijing Academy of Artificial Intelligence (BAAI), Research Intern

Aug. 2024 – Oct. 2025

- Research on embodied-brain benchmarks for multimodal large language models and robot intelligence.

ByteDance, Research Intern

Mar. 2024 – Aug. 2024

- Trust & Safety Applied Algorithm Team for Ads and E-commerce.

HONORS AND AWARDS

- Shanghai Jiao Tong University Outstanding Graduate
- Weichai Power Scholarship
- Huawei Scholarship
- Shanghai Jiao Tong University B-Class Merit Scholarship

Jun. 2023
Oct. 2022
Dec. 2021
2020, 2021, 2022

ACADEMIC SERVICE

- Reviewer for CVPR, NeurIPS, ICLR, AAAI, and ECCV.